CariSurg MedTech Pathways Research Project

Preliminary Proposal Template

Project Title: AI-Assisted Real-Time Emergency Department Triage Decision Support

Student Name: xxx

Statement of the Problem (Tell us, what the problem is):

Emergency departments are experiencing increasing patient volumes and variability in triage decision-making, resulting in delays in identifying critically ill patients. While AI-assisted triage models demonstrate strong predictive performance in retrospective studies, they are predominantly trained on single-centre datasets and lack validation in real-time clinical workflows. This limits their reliability, generalisability, and safe deployment in operational emergency department settings. There is therefore a need for clinician-in-the-loop AI systems that can support consistent triage decisions without disrupting existing clinical workflows.

Previous Work (Tell us what work has been done to solve the problem or a similar problem so far .. based on existing literature (recently within the last 10 years):

Recent literature shows strong and consistent interest in applying artificial intelligence (AI) and machine learning (ML) to improve emergency department triage accuracy, speed, and resource allocation.

Tyler et al. (2024) provide a scoping review of AI applications in ED triage and report improvements in patient prioritisation and predictive accuracy for outcomes such as admission and mortality. However, most included studies are retrospective and lack prospective validation or standardised evaluation frameworks.

Da'Costa et al. (2025) similarly identify benefits such as reduced wait times and improved resource allocation, but highlight key barriers including algorithmic bias, poor data quality, and limited clinician trust, which hinder real-world adoption.

Araouchi & Adda (2024) demonstrate that machine learning models (including SVM, RF, and XGBoost) can predict KTAS triage levels with ~80% accuracy. However, the study is limited by a small, single-country dataset and reduced external validity.

Raita et al. (2020) show that ML models can outperform traditional clinical scoring systems in predicting critical outcomes such as ICU admission and mortality, but emphasise poor external validation and limited clinical implementation in real-time workflows.

Hong et al. (2021) demonstrate improved hospital admission prediction using ML models such as XGBoost and neural networks, but identify dataset bias and lack of integration into live emergency department systems as major limitations.

Jiang et al. (2023) show strong predictive performance of ML models for cardiovascular triage cases, but findings are limited by disease-specific datasets and reduced applicability across general ED populations.

Across all studies, a consistent limitation is the reliance on retrospective datasets and the absence of prospective evaluation in live emergency department workflows.

Proposed Solution (What is in your opinion the way to solve this problem):

This project proposes a 12-week pilot of a clinician-in-the-loop AI-assisted triage decision-support system integrated into emergency department intake workflows.

The system will provide real-time risk stratification and decision support aligned with existing triage scales rather than replacing clinical judgement. It will be designed to operate in a "shadow or assistive mode" to avoid disruption to clinical flow.

The pilot will evaluate whether AI-assisted support improves:

- triage consistency between clinicians
- reduction in under- and over-triage events
- feasibility of real-time integration without increasing triage time

The focus is on prospective workflow evaluation rather than offline model performance.

References (5 to 10 *peer-reviewed journal or conference papers*):
Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Cureus, 16(5), e59906. https://doi.org/10.7759/cureus.59906

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. International Journal of Medical Informatics, 197, 105838. https://doi.org/10.1016/j.ijmedinf.2025.105838

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. Procedia Computer Science, 251, 430–437. https://doi.org/10.1016/j.procs.2024.11.130

Raita, Y., Goto, T., Faridi, M. K., Brown, D. F., & Camargo, C. A. (2020). Emergency department triage prediction of critical illness using machine learning approaches. *Annals of Emergency Medicine*, 76(4), 407–418. https://doi.org/10.1016/j.annemergmed.2020.04.021

Hong, WS., Haimovich, AD., & Taylor, RA. (2021). Prediction of hospital admission at emergency department triage using machine learning. *JAMA Network Open*, 4(6), e2112827. https://doi.org/10.1001/jamanetworkopen.2021.12827